



2003 Assessment Task 2

MATHEMATICS

Extension 2

Year 12

Time allowed - 90 minutes

Topics; Complex Numbers and Polynomials

Instructions

NAME _____

- Attempt all questions.
- Questions are NOT of equal value.
- All necessary working should be shown in every question.
- Marks may be deducted for careless or badly arranged work.
- Write on one side of the paper only.
- Questions do not necessarily appear in order of difficulty.

Question One (27 marks)

1. Given $z = 1 - 3i$ find in the form $a + ib$ (where applicable) [7]

- a) $\bar{z}$ b) z^2 c) $z\bar{z}$ d) $\frac{1}{z}$ e) $z + iz$
 f) $\arg(z)$ g) $\operatorname{Re}(z)$

2. Given $z_1 = a + ib$ and $z_2 = c + id$ prove $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$ [2]

3. Find the square roots of $5 - 12i$ [3]

4. Sketch the locus of the following: [6]

- a) $\operatorname{Im}(z) < 2 \cap \operatorname{Re}(z) < -1$ b) $|z| < 4 \cap \arg(z) \geq \frac{\pi}{4}$

c) $\operatorname{Arg}\left(\frac{z - 2i}{z + 2i}\right) = \frac{\pi}{2}$

5. a) Find the three cube roots of unity and plot them on an Argand diagram. [3]

b) If w is the root with the smallest positive argument show the other is w^2 [1]

c) Show $1 + w + w^2 = 0$ [1]

d) Evaluate $(4 + w + w^2)(2 + 3w + 3w^2)$ [1]

e) Show $\frac{a + bw + cw^2}{b + cw + aw^2} = w$ [1]

f) Form the quadratic equation with roots $aw + bw^2$ and $aw^2 + bw$ [2]

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Question Two (26 marks)

1. a) Write $(-\sqrt{3} + i)$ in modulus argument form [2]
 b) Hence or otherwise evaluate $(-\sqrt{3} + i)^6$ [1]
2. If $z = \cos \theta + i \sin \theta$: [7]
 a) Show that $z^n + \frac{1}{z^n} = 2 \cos n\theta$
~~b)~~
 b) Express in $\cos^4 \theta$ terms of $\cos n\theta$
 c) Find $\int \cos^4 \theta d\theta$
3. a) Sketch the locus of $\arg(z-2) = \frac{2\pi}{3}$ [2]
 b) Hence find z so that $|z|$ is a minimum [2]
4. Find the Cartesian equation of $|z-8| = 3|z-2i|$ [3]
5. a) Solve $z^5 + 1 = 0$. [4]
 b) Hence factorise $z^5 + 1$ over the real field. [3]
 c) Prove that $\cos \frac{\pi}{5} + \cos \frac{3\pi}{5} = \frac{1}{2}$. [2]

Question Three (27 marks)

1. Fully factorise $x^4 - 36$ over the complex field [3]
2. Solve the equation $x^4 - 5x^3 - 9x^2 + 81x - 108 = 0$ given that it has a root of multiplicity three. [4]
3. The polynomial equation $2x^3 + 3x^2 - 4x - 4 = 0$ has roots α, β and γ . [4]
 Find the polynomial equation with roots:
 a) $2\alpha, 2\beta$ and 2γ
 b) α^2, β^2 and γ^2
4. Find the equation of a cubic polynomial that is monic, odd and gives a remainder of 4 when divided by $x+1$ [3]
5. Given the polynomial equation $3x^3 + 5x^2 + 10x - 4 = 0$ [4]
 a) Show that $(-1 + \sqrt{3}i)$ is a root.
 b) Hence or otherwise find the other roots.
6. Given that $px^4 + 4qx + r = 0$ has a double root α : [5]
 a) Show that $p\alpha^3 + q = 0$
 b) Hence prove $27q^4 = pr^3$
7. Given the polynomial $P(x) = x^3 - 6x^2 + 9x + k$ find the values of k , for which $P(x) = 0$ has exactly one root. [4]
 $\uparrow$
 real

Q1 Soln's.

1. $z = 1 - 3i$

a) $\bar{z} = 1 + 3i$

b) $z^2 = (1 - 3i)^2$

$= -8 - 6i$

c) $z\bar{z} = (1 - 3i)(1 + 3i)$
 $= 10$

d) $\frac{1}{1 - 3i} \times \frac{1 + 3i}{1 + 3i} = \frac{1}{10} + \frac{3i}{10}$

e) $z + iz = 1 - 3i + i(1 - 3i)$
 $= 1 - 3i + i - 3$
 $= -2 - 2i$

f) $\arg z = \tan^{-1} \frac{-3}{1}$
 $= -71.07^\circ$
 $(-1.249...)$

g) $\operatorname{Re}(z) = 1$

2. LHS = $\overline{a+ib} + \overline{c+id}$
 $= a - ib + c - id$
 $= (a+c) - i(b+d)$

RHS = $\overline{a+ib+c+id}$
 $= \overline{(a+c) + i(b+d)}$
 $= (a+c) - i(b+d)$
 $= \text{LHS}$

3. let $a+ib = \sqrt{5-12i}$

then $a^2 - b^2 + 2abi = 5 - 12i$

equating Re, Im

$a^2 - b^2 = 5$ (1)

$2ab = -12$ (2)

$(a^2 + b^2)^2 = (a^2 - b^2)^2 + 4a^2b^2$

$= 25 + 144$

$a^2 + b^2 = 13$ (3) $a^2 + b^2 > 0$

(1)+(3) $2a^2 = 18$

$a = 3$

or $a = -3$

$b = -2$

or $b = 2$

ie $3 - 2i$

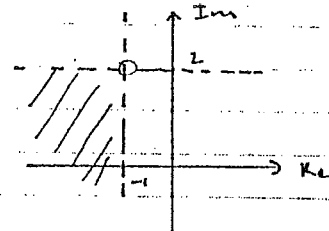
(1)

$-3 + 2i$

(1)

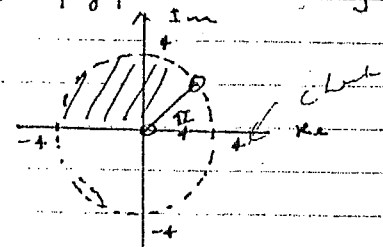
4. a) $\operatorname{Im}(z) < 2 \cap \operatorname{Re}(z) < -1$

ie $y < 2 \cap x < -1$



(2)

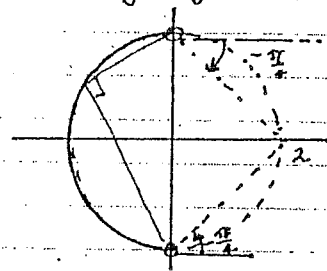
b) $|z| < 4 \cap \arg(z) \geq \frac{\pi}{4}$



(2)

c) $\arg\left(\frac{3-2i}{z+2i}\right) = \frac{\pi}{2}$

ie $\arg(z - 2i) = \arg(z + 2i) = \frac{\pi}{2}$



Check

$-\frac{\pi}{4} - \frac{\pi}{4} = -\frac{\pi}{2}$

ie LHS

Q5. part f) cont

$$\begin{aligned} \Delta B &= (a^2w + bw^2)(aw^2 + bw) \\ &= a^2w^3 + abw^4 + abw^3 + b^2w^4 \\ &= a^2 + abw^4 + abw^3 + b^2 \\ &= a^2 + b^2 + (ab + abw + abw^2 - ab) \\ &= a^2 + b^2 - ab \end{aligned}$$

ie $z^2 + (a+b)z + a^2b - ab = 0$

(2)

5. a) $z^2 = 1$

$$z^0 = \cos 0$$

make general

$$j^2 = c^2 \lambda \pi k \quad \rightarrow \quad k = 0, 1, 2$$

take cube root apply De Moivre's Th

$$3^{\circ} = 2\pi \frac{2\pi k}{3}$$

where $k = 0$

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1-2-2

$$y = 0, 0$$

$$z_1 = 2\sqrt{3} \frac{2\pi}{3}$$

$$z_1 = a; \frac{4\pi}{5} \quad (2)$$

Figure 1

$$\sin C_1 = \frac{L_1}{5}$$

$$b) \quad w = c \cdot \frac{2\pi}{3} \Rightarrow w^2 = c \cdot \frac{4\pi}{3}$$

$$= c_1 - \frac{2\pi}{3}$$

- 3 -

c) $1 + \omega + \omega^2$ are roots of $z^3 - 1 = 0$

Sum of roots = $1 + w + w^2$

$$= -\frac{b}{a}$$

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$$\begin{aligned} d). & (4 + w + w^2) \cdot (2 + 3w + 3w^2) \\ &= (3 + 1 + w + w^2) \cdot (3 + 3w + 3w^2 - 1) \\ &= -3 \end{aligned}$$

$= -3$

e) $\frac{a + bw + cw^2}{b + cw + aw^2} \times \frac{w}{w}$

$$= \frac{a + bw + cw^2}{bw + cw^2 + a} \times w$$

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f) eqn of form

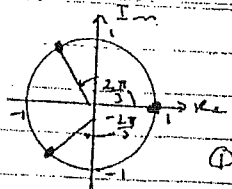
$$z^2 - (\alpha + \beta)z + \alpha\beta = 0$$

$$\alpha + \beta = a_1 v + b_1 v + a_2 v + b_2 v$$

$$= (a + ar + ar^2 + \dots + ar^{n-1}) + (b + br + br^2 + \dots + br^{n-1})$$

$$= -a - b$$

Remainder See end of Q1 part 4)



1- ~~1~~ ② $(-\sqrt{3} + i)$ = $2(-\frac{\sqrt{3}}{2} + \frac{i}{2})$
2 = $2 \operatorname{cis} \frac{5\pi}{6}$

$$\begin{aligned} \textcircled{b}. (-\sqrt{3} + i)^6 &= \left(2 \operatorname{cis} \frac{5\pi}{6} \right)^6 \\ &= 2^6 \operatorname{cis} \frac{5\pi}{6} \times 6 \quad \text{By De Moivre's Thm.} \\ &= 64 \operatorname{cis} 5\pi \\ &= 64 \operatorname{cis} \pi \\ &= 64 \times -1 = -64 \end{aligned}$$

2/ (a) $z = \cos \theta + i \sin \theta$
 $z^n = (\cos \theta + i \sin \theta)^n$
 $= \cos n\theta + i \sin n\theta. \text{ (By De Moivre's Thm)}$

$$z^{-n} = (\cos(-n\theta) + i \sin(-n\theta)) \quad \text{By De Moivre's Thm.}$$

$$\therefore \frac{1}{z^h} = \cosh \theta - i \sinh \theta \quad (\cos \text{ is even, } \sin \text{ is odd})$$

$$\text{Now } z^n + \frac{1}{z^n} = \cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta$$

2// 1b) Now $2 \cos \theta = 3 + \frac{1}{2}$

$$\therefore (2\cos\theta)^4 = \left(\frac{3}{8} + \frac{1}{8}\right)^4$$

$$16 \cos^4 \theta = z^4 + 4z^3 \cdot \frac{1}{z} + 6z^2 \cdot \frac{1}{z^2} + 4z \cdot \frac{1}{z^3} + \frac{1}{z^4}$$

$$2 \quad 16 \cos^4 \theta = z^4 + \frac{1}{z^4} + 4 \left(z^2 + \frac{1}{z^2} \right) + 6$$

$$16 \cos^4 \theta = 2 \cos^4 \theta + 4(2 \cos^2 \theta) + 6$$

$$\therefore \cos^4 \theta = \frac{1}{8} \cos 4\theta + \frac{1}{2} \cos 2\theta + \frac{3}{8}$$

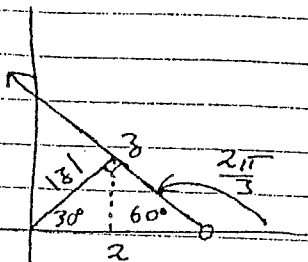
c)
$$J = \int \cos^4 \theta d\theta$$

$$2 = \int_0^{\pi} \left(\frac{1}{8} \cos 4\theta + \frac{1}{2} \cos 2\theta + \frac{3}{8} \right) d\theta$$

$$\eta = \frac{1}{32} \sin 4\theta + \frac{1}{4} \sin 2\theta + \frac{3}{8} \theta + C$$

3/ a)

2



b)

$$\sin 60 = \frac{|\sqrt{3}|}{2}$$

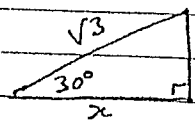
$$\frac{\sqrt{3}}{2} = \frac{|\sqrt{3}|}{2}$$

$|z|$ is a minimum when z is perpendicular

OR arg $z = \frac{\pi}{6}$

2

$$\therefore |z| = \sqrt{3}$$



$$\therefore z = \sqrt{3} \cos \frac{\pi}{6} + i \sqrt{3} \sin \frac{\pi}{6}$$

$$x = \sqrt{3} \cos 30 = \sqrt{3} \cdot \frac{\sqrt{3}}{2} = \frac{3}{2}$$

$$y = \sqrt{3} \sin 30 = \sqrt{3} \cdot \frac{1}{2} = \frac{\sqrt{3}}{2}$$

$$z = \frac{3}{2} + i \frac{\sqrt{3}}{2} = \frac{1}{2}(3 + i\sqrt{3})$$

4/ $|z-8| = 3|z-2i|$

$$\sqrt{(x-8)^2 + y^2} = 3\sqrt{x^2 + (y-2)^2}$$

$$x^2 - 16x + 64 + y^2 = 9(x^2 + y^2 - 4y + 4)$$

$$x^2 - 16x + 64 + y^2 = 9x^2 + 9y^2 - 36y + 36$$

$$64 = 8x^2 + 16x + 8y^2 - 36y + 36$$

$$8 = x^2 + 2x + y^2 - \frac{9}{2}y + \frac{9}{2}$$

$$(x + \frac{1}{2})^2 + 8 = (x^2 + 2x + 1) + y^2 - \frac{9}{2}y + (\frac{9}{2})^2 + \frac{9}{2}$$

$$8 + \frac{81}{16} + 1 - \frac{9}{2} = (x + \frac{1}{2})^2 + (y - \frac{9}{4})^2$$

$$\frac{9}{2} + \frac{81}{16} = \frac{72+81}{16} = (x + \frac{1}{2})^2 + (y - \frac{9}{4})^2$$

$$\frac{153}{16} = (x + \frac{1}{2})^2 + (y - \frac{9}{4})^2$$

Circle: Centre $(-\frac{1}{2}, \frac{9}{4})$

Radius $\frac{\sqrt{153}}{4}$

3/ a)

$$z^5 + 1 = 0 \Rightarrow z^5 = -1 \quad \text{cis } \pi = -1$$

$$z = \text{cis } \theta$$

$$z^5 = (\text{cis } \theta)^5 = \text{cis } 5\theta \quad (\text{by De Moivre's Thm})$$

$$\text{Hence } z^5 = \text{cis } (\pi + 2k\pi)$$

$$\therefore \text{cis } 5\theta = \text{cis } (\pi + 2k\pi)$$

$$\therefore \theta = \frac{\pi + 2k\pi}{5} \text{ for } k=0, 1, 2, 3, 4$$

$$\therefore \theta = \frac{\pi}{5}, \frac{3\pi}{5}, \frac{5\pi}{5} = \pi, \frac{7\pi}{5} = \frac{3\pi}{5}, \frac{9\pi}{5} = \frac{4\pi}{5}$$

$$\therefore \text{Roots of } z^5 + 1 = 0 \text{ are } z = \text{cis } \frac{\pi}{5}, \text{cis } \frac{3\pi}{5}, \text{cis } \frac{5\pi}{5}, \text{cis } \frac{7\pi}{5}, \text{cis } \frac{9\pi}{5}$$

$$\text{Now } z^5 + 1 = (z+1)(z^4 - z^3 + z^2 - z + 1)$$

$$\therefore z^4 - z^3 + z^2 - z + 1 = 0$$

$$\text{Has roots } z = \text{cis } \pm \frac{\pi}{5}, \text{cis } \pm \frac{3\pi}{5}, z = -1$$

b)

$$z^5 + 1 = (z+1)(z^2 - 2\cos \frac{\pi}{5}z + 1)(z^2 - 2\cos \frac{3\pi}{5}z + 1)$$

$$\text{as } z = \text{cis } \frac{\pi}{5} \text{ and } z = \text{cis } \frac{4\pi}{5}$$

are conjugate pairs

$$\therefore (\text{cis } \frac{\pi}{5})(\text{cis } \frac{4\pi}{5}) = 1$$

$$\text{cis } \frac{\pi}{5}z + \text{cis } \frac{4\pi}{5}z = (\cos \frac{\pi}{5} + i \sin \frac{\pi}{5})z + (\cos \frac{4\pi}{5} + i \sin \frac{4\pi}{5})z$$

$$= 2\cos \frac{\pi}{5}z$$

$$\text{Same for } \text{cis } \frac{2\pi}{5}, \text{cis } \frac{3\pi}{5}$$

$$\text{c) Now } z^5 + 1 = (z+1)(z^4 - z^3 + z^2 - z + 1)$$

$$\therefore (z^2 - 2\cos \frac{\pi}{5}z + 1)(z^2 - 2\cos \frac{3\pi}{5}z + 1) = z^4 - z^3 + z^2 - z + 1$$

Equating coefficients of z

$$-2\cos \frac{\pi}{5} - 2\cos \frac{3\pi}{5} = -1$$

$$\therefore \cos \frac{\pi}{5} + \cos \frac{3\pi}{5} = \frac{1}{2}$$

1. $x^4 - 36 = (x^2 - 6)(x^2 + 6)$ Task 2
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 $= (x - \sqrt{6})(x + \sqrt{6})(x - i\sqrt{6})(x + i\sqrt{6})$

2. $f(x) = x^4 - 5x^3 - 9x^2 + 8x - 108$
 $f'(x) = 4x^3 - 15x^2 - 18x + 8$

$f''(x) = 12x^2 - 30x - 18$
 $= 6(2x^2 - 5x - 3)$
 $= 6(2x + 3)(x - 3)$

$f'(-1) \neq 0, f'(3) = 0 \therefore f(x) = (x-3)^3(x+4)$

3. $2x^4 + 3x^2 - 4x - 4 = 0$ has roots α, β, γ

$\therefore (\frac{x}{2} - \alpha)(\frac{x}{2} - \beta)(\frac{x}{2} - \gamma) = 0$ has roots $2\alpha, 2\beta, 2\gamma$

$\Rightarrow 2(\frac{x}{2})^3 + 3(\frac{x}{2})^2 - 4(\frac{x}{2}) - 4 = 0$

$\therefore \frac{x^3}{2} + \frac{3x^2}{2} - 2x - 4 = 0$

OR $x^3 + 3x^2 - 4x - 8 = 0$

$(\sqrt{x} - \alpha)(\sqrt{x} - \beta)(\sqrt{x} - \gamma) = 0$ has roots $\alpha^2, \beta^2, \gamma^2$

$\therefore 2x\sqrt{x} + 3x - 4\sqrt{x} - 4 = 0$

$\therefore 2\sqrt{x}(x-2) = 4-3x$

$\therefore 4x(x^2 - 4x + 4) = 16 - 24x + 9x^2$

$\therefore 4x^3 - 16x^2 + 16x = 9x^2 - 24x + 16$

$\therefore 4x^3 - 25x^2 + 40x - 16 = 0$

4. $f(x) = x^3 + kx$, $f(-1) = 4$

$\therefore 4 = -1 - k \Rightarrow k = -5, \therefore f(x) = x^3 - 5x$

5. a) $3x^3 + 5x^2 + 10x - 4 = 0$

$\sqrt[3]{-1 + i\sqrt{3}}$ is a root, so is $-1 - i\sqrt{3}$

Let $S(x) = (x + 1 + i\sqrt{3})(x + 1 - i\sqrt{3}) = x^2 + 2x + 4$

$$\begin{array}{r} 3x - 1 \\ 3x^2 + 5x^2 + 10x - 4 \\ \underline{3x^3 + 6x^2 + 12x} \\ -x^2 - 2x - 4 \\ \underline{-x^2 - 2x - 4} \\ 0 \end{array}$$

$\therefore f(x) = (3x - 1)(x^2 + 2x + 4)$

has zeros at $\frac{1}{3}, -1 \pm i\sqrt{3}$

6a) $px^4 + 4qx + r = 0$ has a double root α

Let $f(x) = px^4 + 4qx + r$

$f'(x) = 4px^3 + 4q$

$f'(\alpha) = 0 \therefore 4p\alpha^3 + 4q = 0$

$\therefore p\alpha^3 + q = 0 \Rightarrow \alpha^3 = -\frac{q}{p}$

$f(\alpha) = 0, \therefore p\alpha^4 + 4q\alpha + r = 0$

b) $\therefore \alpha(p\alpha^3 + 4q) = -r$

$\therefore \left(-\frac{q}{p}\right)^3 \{-q + 4q\} = -r$

$-\frac{q}{p}(27q^3) = -r^3$

$\therefore 27q^4 = pr^3$

7. $p(x) = x^3 - 6x^2 + 9x + k$

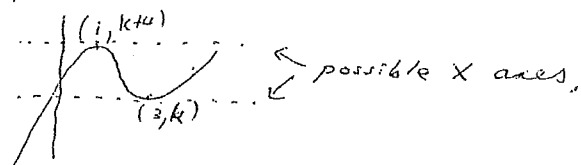
$p'(x) = 3x^2 - 12x + 9$

$= 3(x^2 - 4x + 3)$

$= 3(x-1)(x-3)$

$p(1) = 1 - 6 + 9 + k = k + 4$

$p(3) = 27 - 54 + 27 + k = k$



Now $k+4 < 0$ or $k > 0$

$\therefore k < -4$ or $k > 0$ gives one real root.